

MANAS KUMAR THAKUR

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Education

Bachelor of Technology in Computer Science (AIML)

Expected Graduation: 2027

Dronacharya College of Engineering

CGPA: 7.5/10

Experience

AI Research Intern

Mar. 2026 – May 2026

Daegu Catholic University

Daegu, South Korea – Onsite

- Worked on advanced computer vision and anomaly detection architectures including YOLO, ResNet, CNN-based models, and Amazon PatchCore for industrial inspection systems.
- Designed, trained, and evaluated deep learning pipelines for high-precision defect detection, anomaly localization, and automated quality inspection on industrial datasets.
- Conducted research on next-generation computer vision architectures focused on improving precision, robustness, and reliability for fine-grained industrial anomaly detection tasks.
- Performed model experimentation, optimization, and performance evaluation using Python, PyTorch, OpenCV, and modern deep learning frameworks in real-world industrial environments.

Software Development Intern (AI Engineer)

Jun. 2025 – Aug. 2025

MagicAPI Inc. (Apimarket, Noveum.ai, magicapi)

San Francisco, California, United States – Remote

- Built multi-agent AI systems to automate web content analysis, saving 10+ manual hours/week.
- Built custom tools & microservices, speeding data processing by 25% & improving AI workflow accuracy.
- Contributed to full-stack development, resolving 30+ bugs and testing new application features.

Backend Developer Intern

Feb. 2025 – May 2025

OSS & Consulting

Remote, India

- Designed a 15+ table PostgreSQL database for a new CRM, improving data consistency for client management.
- Collaborated on end-to-end application design, aligning backend logic with CRM features to streamline user experience.
- Deployed and maintained backend infrastructure on a local bare-metal cloud for early-stage product rollout.
- Led backend migration from on-premise to GCP VM instances, improving scalability & reducing latency by 30%.

Technical Projects

IDE 2024 - ML for Construction Progress Monitoring | Deep Learning, OpenCV, Python

September 2024

- Built a real-time ML system to track construction progress via automated image analysis.
- Implemented an anomaly detection module, flagging 98% of reporting errors before they occurred.
- Reduced manual progress tracking time by 30% and significantly improved data accuracy.

SIH 2023 - Lithology Project | YOLOv8, Computer Vision, Python

December 2023

- Developed an ML/CV system for real-time object detection and classification.
- Scraped and annotated a 1,500+ custom lithology image dataset, ensuring high-quality training data.
- Split dataset 70:20:10 for training, validation, and testing, enabling robust model evaluation.
- Fine-tuned a YOLOv8 model, achieving 99.97% accuracy and 99.83% precision on the test set.

Technical Skills

Programming Languages: Python, SQL, C++, JavaScript, R

AI & Machine Learning: Deep Learning, Generative AI, Agentic AI Systems, Multi-Agent Workflows, Computer Vision, NLP, Reinforcement Learning, RAG, Fine-Tuning, Anomaly Detection

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, OpenCV, Scikit-learn, FastAPI, YOLOv8

Model Architectures: Transformers, Vision Transformers (ViTs), CNNs, LSTMs, U-Net, CLIP, DINO, ResNet, PatchCore, SAM

LLM & Agentic Systems: Tool Calling, Function Calling, Prompt Engineering, Context Management, Memory Systems, Workflow Orchestration, Autonomous Agents, Vector Retrieval Pipelines

Data Science & Visualization: Pandas, NumPy, Matplotlib, Plotly, Streamlit, Jupyter Notebook

MLOps & Infrastructure: Docker, Git, CI/CD, MLflow, Weights & Biases, CUDA, NVIDIA Jetson, Linux, GCP, REST APIs

Databases & Vector Stores: PostgreSQL, MongoDB, Redis, Pinecone, LanceDB, ChromaDB

Leadership & Achievements

Smart India Hackathon Winner: SIH 2023 Winner for automated lithology classification using AI/DL.

Engineers Day Idea Competition (DCE): 2nd place for designing a fire-resistant system for car accidents.

IDE Bootcamp Selection: Advanced to Phase 2 at Amity University, collaborating on innovations.

Technical Lead: Currently leading technical projects and workshops at Google Developer Group (GDG) Club.

AI Team - Deviators Club: Gained hands-on AI/ML project experience in advanced technologies.